$$\lambda_i=(\pi i)^2/2\pm[(\pi i)^4/4+(\pi i)^2]^{0.5}, \qquad i=1, 2, \ldots$$

In conclusion, I would like to thank A.V. Gulin for his interest in this research.

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THE MULTIDIMENSIONAL STATIONARY PHASE METHOD. THE SECOND TERM OF THE ASYMPTOTIC EXPANSIONS*

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The second term in the asymptotic expansion of the contribution from a nondegenerate stationary point is determined for the multidimensional case.

Consider the integral

$$F(\lambda)=\int_{\mathbb{R}^n} f(x)\exp[i\lambda S(x)]\,dx, \tag{1}$$

where $f\in C_0^\infty(\mathbb{R}^n)$, $S\in C^\infty(\mathbb{R}^n)$, S is a real-valued function, and $\lambda>0$ is a large parameter. Assume that S has precisely one stationary point x^0 on supp f. The point x^0 is said to be nondegenerate if

$$\det H(x^0)\neq 0, \qquad H(x^0)=S_{xx}''(x^0). \tag{2}$$

If the point x^0 is nondegenerate, we have the asymptotic expansion as $\lambda\to+\infty$ [1]

$$F(\lambda)\sim c_n(x^0)(2\pi/\lambda)^{n/2}\exp[i\lambda S(x^0)]\left[f(x^0)+\sum_{m=1}^{\infty}a_m\lambda^{-m}\right], \tag{3}$$

$$c_n(x^0)=|\det H(x^0)|^{-1/2}\exp\left[i\frac{\pi}{4}\operatorname{sgn}H(x^0)\right],$$

where $\operatorname{sgn}H(x^0)$ is the difference between the number of positive and negative eigenvalues of the matrix $H(x^0)$.

The main result of this paper is the formula

$$a_1=iLf-\frac{i}{2}[fL^2S+2L(f,LS)], \tag{4}$$

where

$$Lf={}^1/_2(H^{-1}(x^0)\nabla,\nabla)f, \qquad L(f,\varphi)=(H^{-1}(x^0)\nabla f,\nabla\varphi). \tag{5}$$

In (4), all the functions and their derivatives are evaluated at the point x^0. Let us prove (4). Introduce the function

$$S(x,x^0)=S(x)-S(x^0)-{}^1/_2(H(x^0)(x-x^0),\ x-x^0)$$

and apply formula (2.13) from [1, p.189] (note that the factor ½ has been omitted in [1]). For brevity, let $x^0=0$, $S(x^0)=0$, $S(x,x^0)=S(x)$, $H^{-1}(x^0)=A$ and put $f_j=\partial f/\partial x_j$, $f_{jk}=\partial^2 f/\partial x_j\partial x_k$ etc. We have [2]

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$$f(0)+\sum_{m=1}^{\infty} a_m\lambda^{-m}=\sum_{m=0}^{\infty}\frac{1}{m!}\left(\frac{i}{\lambda}\right)^m L^m(f\exp(i\lambda S))|_{x=0}. \tag{6}$$

The function $S(x)$ has a zero of order not less than 3 at the point $x = 0$, and therefore a_1 is equal to the coefficient of λ^{-1} in the sum

$$\frac{i}{\lambda}L(f\exp(i\lambda S))-\frac{1}{2\lambda^2}L^2(f\exp(i\lambda S)) \tag{7}$$

for $x = 0$. From (5) we have

$$L(f\exp(i\lambda S))=\exp(i\lambda S)\frac{1}{2}a_{jk}M_{jk}(f), \tag{8}$$
$$M_{jk}(f)=f_{jk}+i\lambda(f_jS_k+f_kS_j+fS_{jk})-\lambda^2 fS_jS_k,$$

where $A = (a_{jk})$ and summation over similar indices is assumed. For $x = 0$ we have $S = 0$, $S_j = 0$, $S_{jk} = 0$, so that $(M_{jk}f)(0)=f_{jk}(0)$, $L(f\exp(i\lambda S))|_{x=0}=Lf|_{x=0}$, and the coefficient of λ^{-1} in the first term in (7) is $iLf|_{x=0}$ It remains to compute the coefficient b of λ in the expression $L^2(f\exp(i\lambda S))|_{x=0}$. By (8)

$$b=\frac{i}{4}a_{jk}a_{lm}[f_{jklm}+f_jS_{klm}+f_kS_{jlm}+f_mS_{jkl}+f_lS_{jkm}+fS_{jklm}]=$$
$$=i[fL^2S+2L(f,LS)],$$

where all functions are evaluated at the point $x = 0$. This proves (4).

Expansion (3) can be viewed from a different angle. (We will only give the formal algebra; a rigorous proof can be obtained using the results of [2].) we have

$$F(\lambda)=\int_{-\infty}^{\infty}\exp(i\lambda c)\Phi(c)\,dc,\qquad \Phi(c)=\int_{\mathbb{R}^n}f(x)\delta(S(x)-c)\,dx. \tag{9}$$

Let $S(0) = 0$ and let $x = 0$ be a nondegenerate stationary point. It is natural to try and expand the function $\delta(S-c)$ in a Taylor series in powers of c, but because $\nabla S(0)=0$ we obtain a series in powers of $c^{1/2}$. Represent S in the neighbourhood of the point $x = 0$ in the form

$$S(x)=S_2(x)+S_3(x)+\ldots, \tag{10}$$

where $S_j(x)$ is a homogeneous polynomial of degree j, and make the change of variables $x = c^{1/2}y$. Then from (9) and (10) we obtain

$$F(\lambda)=\int_{-\infty}^{\infty}c^{-n/2}\exp(i\lambda c)\Psi(c)\,dc,\quad \Psi(c)=\int\chi(y,c)\,dy, \tag{11}$$
$$\chi(y,c)=\delta(S_2(y)-1+c^{1/2}S_3(y)+\ldots)f(c^{1/2}y).$$

Expanding the function $\chi(y,c)$ in a Taylor series in powers of $c^{1/2}$, we obtain

$$f(c^{1/2}y)=f_0+c^{1/2}f_1(y)+cf_2(y)+\ldots,$$
$$\chi(y,c)=f_0\delta(a)+c^{1/2}[f_0S_3\delta'(a)+f_1\delta(a)]+$$
$$+c[f_2\delta(a)+f_1S_3\delta'(a)+f_0S_4\delta'(a)+{}^1/_2f_0S_3{}^2\delta''(a)]+\ldots.$$

Here $f_j(y)$ is a homogeneous polynomial of degree j, $f_0=f(0)$, $a(y)=S_2(y)-1$. Since the function $a(y)$ is even, and the functions $f_1(y)$ and $S_3(y)$ are odd, the coefficient of $c^{1/2}$ in the expansion of Ψ is zero. We can show that Ψ has a series expansion in integral powers of c. Thus,

$$a_1=\lambda\int_{-\infty}^{\infty}c^{-n/2+1}\exp(i\lambda c)\,dc\int\Big[f_2\delta(a)+f_1S_3\delta'(a)+$$
$$+f_0S_4\delta'(a)+\frac{1}{2}S_3{}^2\delta''(a)\Big]dy.$$

Using the definitions of the function $\delta(a)$ and its derivatives from [2], we obtain (4) from the last formula.

Consider the hypersurface $M^n\subset\mathbb{R}^{n+1}$, defined by the equation $x_{n+1}=S(x)$. The eigenvalues $k_1,\ldots,k_n$ of the matrix $H(x^0)$ are the principal curvatures of M^n at the point x^0, $\Delta(x^0)=K$ is the total curvature, and

$$|\Delta(x^0)|^{-1/2}\exp\left(i\frac{\pi}{4}\operatorname{sgn}H(x^0)\right)=K^{-1/2}$$

with an appropriate choice of the branch of the square root.

Consider the example

$$S(x)=\frac{1}{2}\sum_{j=1}^{n}k_jx_j^2,$$

then

$$L=\frac{1}{2}\sum_{j=1}^{n}k_j^{-1}\frac{\partial^2}{\partial x_j^2},\qquad LS=\frac{n}{2},\quad L^2S=0,\quad L(f,LS)=0,$$

so that

$$a_1=\frac{i}{2}\sum_{j=1}^{n}k_j^{-1}\frac{\partial^2 f}{\partial x_j^2}\bigg|_{x=0}.$$

In this case, we can write out the entire asymptotic series:

$$\int_{\mathbb{R}^n}\exp\Big(\frac{i\lambda}{2}\sum_{j=1}^{n}k_jx_j^2\Big)f(x)\,dx=\Big(\frac{2\pi}{\lambda}\Big)^{n/2}|k_1\ldots k_n|^{-1/2}\times$$
$$\times\exp\Big(i\frac{\pi}{4}\sum_{j=1}^{n}\operatorname{sgn}k_j\Big)\Big[f(0)+\sum_{m=1}^{\infty}\frac{1}{m!}\Big(\frac{i}{\lambda}\Big)^m(L^mf)(0)\Big].$$

The preceding results are applicable to integrals of the form

$$F(\lambda)=\int_{M^n}\exp(i\lambda S)\,\omega,$$

where M^n is an n-dimensional manifold of class C^∞ S $M^n\to\mathbb{R}$ and ω is a differential form on M^n. Consider the important special case

$$F(\lambda)=\int_{g=0}f\exp\{i\lambda S\}\,d\sigma,\tag{12}$$

where $g(x)\in C^\infty(\mathbb{R}^n)$, $n\geqslant 2$, the set M is such that $g = 0$ is nonempty and $\nabla g(x)\neq 0$ on M. Then M is a hypersurface of class C^∞ in $\mathbb{R}^n$, and $d\sigma$ is an area element of this hypersurface. It is shown in [2] that $d\sigma=\alpha(x)\omega$, where ω is the Leray-Gel'fand differential form associated with the function g, so that $F(\lambda)=(\delta(g),\,\alpha f\exp\{i\lambda S\})$. Let $g(0) = 0$ and let $x = 0$ be a nondegenerate stationary point of the restriction of the function S on M. Then

$$\nabla S(0)=\mu\nabla g(0),\qquad \mu\neq 0\tag{13}$$

In a small neighbourhood U of the point $x = 0$,

$$g(x)=(x,a)+\frac{1}{2!}g_2(x)+\ldots,\qquad S(x)=\mu(x,a)+\frac{1}{2!}S_2(x)+\ldots,\tag{14}$$

where g_j, S_j are homogeneous polynomials of degree j. Introduce the Cartesian coordinates $\xi_1,\ \ ,\xi_n$ $\xi'=(\xi_1,\ \ ,\xi_{n-1})$ in U such that the axis ξ_n is parallel to the vector $a=\nabla g(0)$ and the point $\xi=0$ corresponds to the point $x = 0$. Then $(x,a)=|a|\xi_n$. The restriction $S|_M$ is a function of ξ'. An operator analogous to L is defined by the linear and quadratic terms g_1, g_2, S_1, S_2 of the functions g and S. Let

$$g(x)=\tilde g(\xi)=|a|\xi_n+{}^1/_2\tilde g_2(\xi',\xi_n)+\ldots,$$
$$S(x)=\tilde S(\xi)=\mu|a|\xi_n+{}^1/_2\tilde S(\xi',\xi_n)+\ldots,$$

so that

$$S|_M=\frac{1}{2!}S_{20}(\xi')+\frac{1}{3!}S_{30}(\xi')+\ \ldots,\tag{15}$$
$$S_{20}(\xi')=-\frac{\nabla S(0)}{\nabla g(0)}\tilde g_2(\xi',0)+\tilde S_2(\xi',0)=(B\xi',\xi')$$

For the integral (12) we have the asymptotic expansion (3) in which n should be replaced by $n - 1$ and the matrix $H(x^0)$ by the matrix B (see (15)). Moreover, $fd\sigma$ should be represented in the form $f(\xi')d\xi'$ In particular, if g has the form

$$g(x)=ax_n+\frac{1}{2!}g_2(x)+\ldots,$$

then $\xi=x$ and

$$B=-\frac{\nabla S(0)}{\nabla g(0)}g_2(x',0)+S_2(x',0).$$

Similar formulas hold for the Laplace integrals

$$G(\lambda)=\int_{\mathbb{R}^n} f(x)\exp[\lambda S(x)]\,dx.$$

The conditions on f and S are the same as in (1). Let

$$\max_{x\in \operatorname{supp} f} S(x)=S(x^0)$$

and let x^0 be a nondegenerate maximum point. Then as $\lambda\to+\infty$

$$G(\lambda)=\left(\frac{2\pi}{\lambda}\right)^{n/2}|\det H(x^0)|^{-1/2}\exp[\lambda S(x^0)]\left[f(x^0)+\sum_{m=1}^{\infty}b_m\lambda^{-m}\right].$$

This formula is obtained from (3) and (4) if we replace $i\lambda$ by λ, $b_1=-Lf+{}^1/_2[fL^2S+2L(f,LS)]$.

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A DECOMPOSITION APPROACH TO THE SOLUTION OF SOME INTEGER PROGRAMMING PROBLEMS WITH INEXACT DATA*

V.A. ROSHCHIN, N.V. SEMENOVA and I.V. SERGIYENKO

Integer programming problems with inexact data that model various situations are considered. An approach to solving some classes of such problems is proposed.

1. We investigate the integer programming problems

$$\max\{c^Tx\,|\,Ax\leqslant b,\ x\in Z_+^n\},\tag{1}$$

where Z_+^n is the space of n-dimensional vectors with nonnegative integer components, in which all the coefficients are defined by their set memberships (the exact value of the coefficient is unknown). Specifically, we assume that $c\in C$ and $b\in B$, where C and B are convex polyhedra in $\mathbb{R}^n$ ($\mathbb{R}^n$ is an n-dimensional Euclidean space) and $\mathbb{R}^m$, respectively, $A=[a_{ij}]_{m,n}\in H$, $a_{ij}\in\mathbb{R}^1$, and H is a closed set in $\mathbb{R}^{m\times n}$ Problems of this kind arise in the context of mathematical modelling of inexact data, when analyzing the stability of the solutions of such models.

Some classes of mathematical programming problems with inexact parameters are investigated in [1-11] and elsewhere. Linear programming problems in which the constraint matrices may take any value from a given set are considered in [3]. Linear programming problems with interval coefficients are studied in [4]. Some linear programming problems with inexact data are also investigated in [5, 6]. Linear programming problems with approximately specified constraint matrices and right-hand sides are studied in [7]. A general approach to systems of linear equations and inequalities whose coefficients and right-hand sides are defined by set memberships is proposed in [8]. Solution of integer programming problems in which the objective function coefficients take values from a given set of vectors is considered in [9]. An approach to the solution of integer programming

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